

Mohamed Aziz Ben Romdhane

AI Engineer | Machine Learning Engineer

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PROFILE

AI Engineer and Machine Learning Engineer building production-grade Generative AI applications, agentic AI systems, retrieval-augmented generation (RAG) pipelines and Python APIs. Shipped an AI travel concierge with 27 agent tools and 28 streaming endpoints, reducing trip-plan generation time by 65%, and an LLM contract-extraction platform backed by Oracle. Experienced in Python, FastAPI, LangGraph, LangChain, PyTorch, Hugging Face, vector databases, Docker, automated testing and CI/CD, plus NLP, time-series forecasting, anomaly detection and graph neural networks.

EDUCATION

Engineering Degree in Data Science (Bac+5, Master's equivalent)

Sep 2021 – Dec 2025

ESPRIT, École Supérieure Privée d'Ingénierie et de Technologies, Tunis

PROFESSIONAL EXPERIENCE

AI / Software Engineer, Worldsoft

Feb 2026 – Present

- Shipped the **AI travel concierge of the CurvaTrip app: 27 agent tools** and 28 streaming endpoints on FastAPI, live GDS flight search as structured fare cards, trip planning and real-time voice (Gemini Live) in **4 languages** including Tunisian Derja; prompt engineering and cached catalogs **cut plan build from 128 s to 44 s (-65%)**.
- Built an **LLM contract-extraction SaaS** end to end: a Gemini extractor turns supplier PDF/DOCX contracts into a normalised 17-table Oracle schema and a deterministic engine prices any stay, **replacing manual contract entry**. 49 REST endpoints, async job queue, multi-tenant JWT with SSO, React UI, **pytest extraction-quality and pricing suites in GitLab CI/CD**.
- Delivered an **8-agent analytics dashboard** (SQL, analyst, forecast, anomaly, segmentation, hotel, flight, orchestrator) on LangGraph and DeepSeek: **52 REST endpoints** with SSE streaming over Oracle and an Angular 17 UI, replacing hand-built SQL reports with self-service KPIs, charts and maps.

Technologies: Python, FastAPI, LangGraph, LangChain, Gemini, DeepSeek, Oracle, pytest, Angular, React, Flutter, Docker, GitLab CI/CD.

.NET Developer Intern, Prosper Us

Jul 2023 – Aug 2023

- Developed PharmaReport, an ASP.NET Core MVC application generating structured pharmacy reports: built the MedicalReport and ReportSheet modules on EF Core migrations, replacing hand-written report documents. **Technologies:** C#, ASP.NET Core, Razor Pages, EF Core, SQL.

Web Developer Intern, AS2E: client WordPress sites and recurring internal reporting automated with Google Apps Script.

Jul 2022 – Aug 2022

Technologies: WordPress, PHP, JavaScript.

PROJECTS

Flight Invoice RAG Chatbot (personal) | GitHub

May 2026

- Built a **RAG** application that extracts and indexes flight-invoice content with Hugging Face embeddings in a ChromaDB vector store, answering natural-language questions through Groq Llama 3 behind a Streamlit interface.

Knowledge Graph for Project Management (ESPRIT) | GitHub

Sep 2024 – Jan 2025

- Trained a PyTorch **RGCN** for link prediction over a project knowledge graph, ranking missing task, resource and dependency links on a held-out edge split; BERT embeddings for semantic text, Pinecone vector search and GPT-4o mini for natural-language question answering. **Technologies:** Python, PyTorch, RGCN, BERT, Hugging Face, Pinecone.

BVMT Analytics Platform, Tunis Stock Exchange (ESPRIT) | GitHub

Feb 2024 – Jun 2024

- Built a **10-module** Streamlit platform on 8 years of market data (2016-2023): price-direction prediction (AdaBoost, Gradient Boosting), risk classification, DBSCAN + UMAP clustering, Prophet dividend forecasts and a **RAG chatbot** (LangChain, Llama 2, FAISS) over uploaded market PDFs. **Technologies:** Python, Streamlit, Scikit-learn, Prophet, UMAP, LangChain, FAISS, MongoDB.

TECHNICAL SKILLS

Artificial Intelligence & Machine Learning: Generative AI, Agentic AI, Large Language Models (LLMs), retrieval-augmented generation (RAG), LangGraph, LangChain, multi-agent orchestration, prompt engineering, NLP, Transformers, BERT, Hugging Face, PyTorch, Scikit-learn, Pandas, NumPy, graph neural networks, time-series forecasting, model evaluation | **Programming:** Python, SQL / PL-SQL, Java, C#, TypeScript, Bash | **Backend, Model Deployment & APIs:** FastAPI, model serving, REST APIs, Spring Boot, ASP.NET Core, JWT, SSE, OpenAPI | **MLOps & Software Engineering:** Docker, GitLab CI/CD, automated testing, pytest, Git, Linux, Prometheus, Grafana, logging and monitoring | **Data & Vector Stores:** Oracle, MySQL, MongoDB, FAISS, ChromaDB, Pinecone

CERTIFICATIONS

- Building Transformer-Based Natural Language Processing Applications, NVIDIA, 2024
- Natural Language Processing Specialization (4 courses), DeepLearning.AI on Coursera, 2024
- Microsoft Azure fundamentals coursework (3 courses), Microsoft on Coursera, 2024

LANGUAGES

Arabic: Native | **French:** Fluent | **English:** Advanced